

Assignment-2

Implement Scaled Dot-Product Attention and a Mini Transformer Encoder

You are required to implement the core components of the Transformer architecture using PyTorch. The assignment focuses on understanding how **attention** works inside a Transformer.

Problem Statement

Implement a simple **Transformer Encoder** from scratch and test it on a small text classification task.

You should implement the following:

1. **Scaled Dot-Product Attention**
2. **Multi-Head Self-Attention**
3. **Positional Encoding**
4. **Feed Forward Network**
5. **Transformer Encoder Block**
6. **A simple classifier using the Transformer Encoder**

Dataset

Use any small dataset, such as:

- IMDB small subset
- AG News subset
- SMS Spam Detection dataset
- A custom dataset with 2 classes, for example positive/negative sentences

You may also create a small toy dataset manually.

Tasks

1. Preprocess the text data:
 - Tokenize sentences
 - Convert words into integer IDs
 - Pad sequences to the same length

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2. Implement **Scaled Dot-Product Attention** using the formula:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

3. Implement **Multi-Head Attention** by splitting the input into multiple heads.
4. Add **Positional Encoding** to preserve word order information.
5. Build one **Transformer Encoder Block** using:
 - Multi-head self-attention
 - Feed-forward layer
 - Residual connection
 - Layer normalization
6. Train the model for text classification.
7. Report:
 - Training loss
 - Accuracy
 - Example predictions
 - A short explanation of how attention helps the model

Expected Output

Your submission should include:

- Colab Notebook
- Model architecture summary
- Training and testing results
- plots of loss/accuracy
- Brief explanation of each implemented component

Note: No need to make report for this assignment. You can show analysis in colab notebook itself.